

PROJECT WALKTHROUGH

VanillaCart

Function by Function

Every function in the shopping-cart project explained — what it does, how it works, the money pipeline, and the flow diagrams that connect them. One file, four sections.

"A project you can't explain function-by-function is a project you copied. This document is the proof you didn't."

1

JS FILE

18

FUNCTIONS

6

DIAGRAMS

0

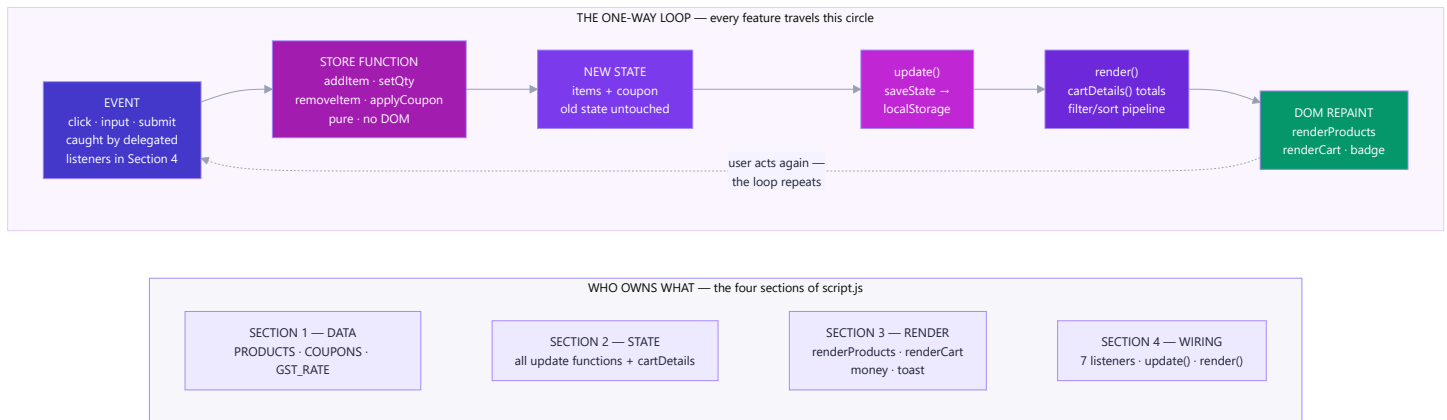
DEPENDENCIES

Table of Contents

- 1. The Skeleton — One Overall Diagram
 - 2. Section 1 — The Data
 - 3. Section 2 — The State Engine, Function by Function
 - 4. Section 3 — The Render Layer, Function by Function
 - 5. Section 4 — The Wiring, Function by Function
 - 6. The Flows — What Happens When...
-

1. The Skeleton — One Overall Diagram

One file — `script.js` — with four labelled sections, four jobs, one direction of data flow. Every feature in the app is some path through this picture:



The rule the diagram encodes: events never touch the DOM directly, and render functions never make decisions. Everything goes *around* the loop: **event** → **store** → **new state** → **save** → **render** → **DOM**. One way, every time.

Telugu: Okate file, naalugu sections — naalugu panulu: **products** = data, **store** = logic (DOM muttadu), **ui** = drawing (decisions undavu), **app** = wiring. Prathi feature ee circle lo ne tirugutundi: event → store → kotha state → save → render → DOM. **Okate direction, prathi saari.**

2. Section 1 — The Data

No functions here — three exported constants, and one design decision.

Export	What it is
PRODUCTS	Array of { id, name, price, category, emoji } — the catalogue
COUPONS	{ SAVE10: 0.10, WELCOME20: 0.20 } — code → fraction off
GST_RATE	0.18 — one named constant instead of a magic number in the maths

The design decision: the cart state stores only { id, qty } — never the name or price. Product facts live in ONE place, so a price change edits one line of one file, and the cart can never show a stale price. `cartDetails()` joins the two at read time, exactly like a SQL JOIN between an `orders` table and a `products` table.

Telugu: Cart lo product **id + qty matrame** dachutam — peru, price kaadu. Price okka chota ne untundi (PRODUCTS lo); cart eppudu paatha price chupinchadu. Idi SQL JOIN laantidi: rendu tables, read time lo kaluputam.

3. Section 2 — The State Engine, Function by Function

Every function here takes state in, returns **new** state out. None of them touch the DOM, none of them mutate their input.

3.1 `emptyState()`

```
const emptyState = () => ({ items: [], coupon: null });
```

The shape of "nothing in the cart," written once and reused by `loadState` and `clearCart`. Note the `({ ... })` — the parentheses make the arrow return an object instead of reading `{}` as a code block (a classic arrow-function trap).

3.2 `loadState()`

```
function loadState() {  
  try {  
    return JSON.parse(localStorage.getItem(STORAGE_KEY)) ?? emptyState();  
  } catch {  
    return emptyState();  
  }  
}
```

How it works, line by line:

1. `localStorage.getItem` returns the saved **string**, or `null` on a first visit.
2. `JSON.parse` turns the string back into a live object. (`JSON.parse(null)` happens to return `null`, so the first-visit case flows through safely.)
3. `?? emptyState()` — if the result was `null` / `undefined`, start fresh. `??` and not `||`, so a legitimate falsy value would survive.
4. `try/catch` — if someone hand-edited the storage into invalid JSON, `parse` throws; a storage problem must never crash the app, so we recover with an empty cart.

Called from: the top of the WIRING section, once, at page load.

Telugu: localStorage **strings matrame** dachutundi — anduke `JSON.parse` tho object ga maarchali. Modatisari visit lo `null` vastundi → `??` khaali cart istundi. Data corrupt aithe `try/catch` app ni crash avvakunda kaapadutundi.

3.3 `saveState(state)`

```
function saveState(state) {  
  localStorage.setItem(STORAGE_KEY, JSON.stringify(state));  
}
```

The mirror of `loadState`: `object` → `JSON.stringify` → `string` → `storage`. The key `"vanillacart-v1"` is versioned on purpose — if a future version changes the state shape, bumping to `-v2` abandons old incompatible data instead of crashing on it.

Called from: `update()` in the WIRING section — which means *every* state change is persisted automatically. There is no way to change the cart and forget to save it.

3.4 `addItem(state, id)`

```
function addItem(state, id) {
  const existing = state.items.find(it => it.id === id);
  const items = existing
    ? state.items.map(it => it.id === id ? { ...it, qty: it.qty + 1 } : it)
    : [...state.items, { id, qty: 1 }];
  return { ...state, items };
}
```

How it works:

1. `find` checks whether this product already has a cart line.
2. **Already there** → `map` builds a *new* array; only the matching line is replaced with a *new* object with `qty + 1`; every other line is reused as-is.
3. **Not there** → spread the old array into a new one and append `{ id, qty: 1 }`.
4. Return a new state object with the new items array. The original `state` is untouched — you could still compare "before vs after," which is exactly how undo features and React change-detection work.

Why never `existing.qty++`? Mutation would change the old state too, killing comparability — and in React it silently breaks re-rendering. The habit you build here is the habit React assumes.

Telugu: Item already unte → `map` tho **kotha array**, aa okka line matrame kotha object (`qty + 1`). Lekapothe → `[...items, {id, qty: 1}]`. **Paatha state ni eppudu muttamu** — anduke prathi function kotha state ni return chestundi. Ide React lo `setState` alavatu.

3.5 `setQty(state, id, qty)`

```
function setQty(state, id, qty) {
  if (qty < 1) return removeItem(state, id);
  const capped = Math.min(qty, 99);
  return {
    ...state,
    items: state.items.map(it => it.id === id ? { ...it, qty: capped } : it),
  };
}
```

How it works:

1. `qty < 1` → delegate to `removeItem`. This one rule means the – button needs no special "was it the last one?" logic anywhere else.
2. `Math.min(qty, 99)` caps the quantity — defensive coding against a rogue value.
3. Same `map` -with-replacement pattern as `addItem`.

Called from: the cart's + and – buttons (`inc` / `dec` actions in the WIRING section).

3.6 `removeItem(state, id)`

```
function removeItem(state, id) {  
  return { ...state, items: state.items.filter(it => it.id !== id) };  
}
```

`filter` keeps every line whose id *doesn't* match — which is how you "delete" immutably: build a new array without the item. One line, no index hunting, no `splice`.

3.7 `clearCart()`

```
function clearCart() { return emptyState(); }
```

Doesn't even need the old state — clearing is just "return the empty shape." The coupon resets too, because the empty shape says `coupon: null`.

3.8 `applyCoupon(state, code)`

```
function applyCoupon(state, code) {  
  const clean = code.trim().toUpperCase();  
  if (!COUPONS[clean]) return { state, ok: false };  
  return { state: { ...state, coupon: clean }, ok: true };  
}
```

How it works:

1. Normalise the input — `" save10 "` → `"SAVE10"`. Never trust raw user input, even for casing.
2. Look the code up in the `COUPONS` table. Unknown code → return the state **unchanged** plus `ok: false`.
3. Known code → new state with `coupon` set, plus `ok: true`.

Why return `{ state, ok }` instead of just state? The caller (the WIRING section) needs to know whether to toast "applied 🎉" or "invalid ❌". Returning both keeps the store silent about UI — it reports facts; the wiring decides what to show.

Telugu: Input ni modata **shubhram cheyyi** (trim + uppercase). Code table lo unte kotha state + `ok: true` ; lekapothe paatha state + `ok: false` . Store UI gurinchi em telidu — **facts matrame cheptundi**, toast em chupinchalo the WIRING section decide chestundi.

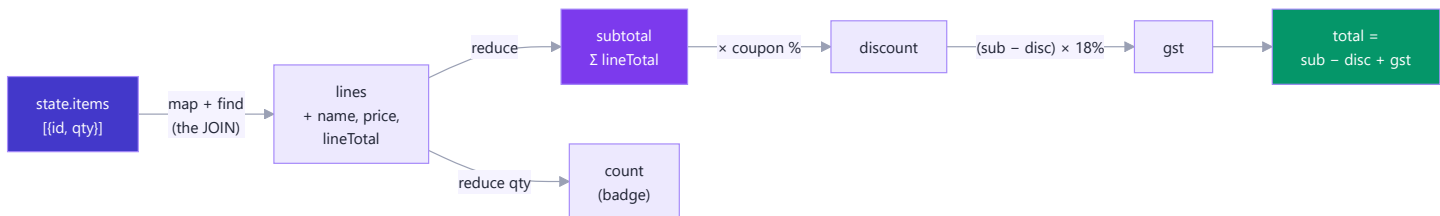
3.9 `cartDetails(state)` — where the money is computed

```
function cartDetails(state) {
  const lines = state.items.map(it => {
    const product = PRODUCTS.find(p => p.id === it.id);
    return { ...product, qty: it.qty, lineTotal: product.price * it.qty };
  });

  const subtotal = lines.reduce((sum, l) => sum + l.lineTotal, 0);
  const discount = subtotal * (COUPONS[state.coupon] ?? 0);
  const gst      = (subtotal - discount) * GST_RATE;
  const total    = subtotal - discount + gst;
  const count    = lines.reduce((sum, l) => sum + l.qty, 0);

  return { lines, subtotal, discount, gst, total, count, coupon: state.coupon };
}
```

This is the money pipeline — five steps, each feeding the next:



How it works:

1. **The JOIN** — each `{ id, qty }` line is enriched with its product's name/price/emoji, plus a computed `lineTotal = price × qty`.
2. **subtotal** — `reduce` folds all lineTotals into one number: start at `0`, add each line.
3. **discount** — `COUPONS[state.coupon] ?? 0` reads the coupon's fraction; no coupon → `0`, so the same formula works with and without one (no if/else).
4. **gst** — 18% of the *discounted* amount (tax applies after discount — a real invoice rule).
5. **total and count** — the grand total, and the badge number (sum of quantities, not number of lines: 2 mice + 1 keyboard = 3).

Why compute instead of store? A stored total goes stale the moment any update path forgets to refresh it. A computed total *cannot* be wrong. This is the "derived data" principle — the single most transferable idea in the project.

Telugu: Money pipeline: lines (JOIN) → subtotal (`reduce`) → discount (coupon %) → GST (discount taruvata 18%) → total. Totals ni **epudu dachamu, prathi saari lekkistham** — dachina total okka update marchipothe tappu aipotundi; lekkinchina total tappu avvadam *saadhyam kaadu*. Idi ee project lo atyanta mukhyamaina idea.

4. Section 3 — The Render Layer, Function by Function

4.1 money(n)

```
const money = n =>
  new Intl.NumberFormat("en-IN", { style: "currency", currency: "INR", maximumFractionDigits: 0 })
    .format(Math.round(n));
```

Turns `14999` into `"₹14,999"` — Indian digit grouping, ₹ symbol, no decimals. `Intl.NumberFormat` is built into the browser: no library, correct lakh/crore-style grouping (`1,00,000`), and swapping `"en-IN"` / `"INR"` for another locale is a one-line internationalisation.

Note the display-rounding subtlety: each shown number is rounded *independently* for display, but `total` is computed from the exact unrounded values. So ₹799 – ₹80 + ₹129 can display alongside a total of ₹849 (the exact maths is 848.54). Real invoices work the same way — round late, not early.

4.2 renderProducts(products)

```
grid.innerHTML = products.map(p => `
  <article class="card">
    ...
    <button class="add" data-id="${p.id}">Add to Cart</button>
  </article>
`).join("");
```

How it works:

1. Takes the already-filtered/sorted array (the WIRING section decides *what*; this function only draws).
2. Empty array → a friendly "no products match" message and an early return.
3. `map` turns each product into an HTML string; `join("")` glues them; **one** `innerHTML` assignment paints the lot. One assignment = one reflow (cheap), versus one reflow per card with repeated `appendChild`.
4. `data-id="${p.id}"` stamps each button with its product's id — this is the bridge the click handler will read back.

Safety note: template-literal HTML is fine here because product names are *our* data. The moment strings come from users, this pattern becomes an XSS hole and you switch to `createElement` + `textContent`.

Telugu: Prathi product ni HTML string ga maarchi (`map`), anni kalipi (`join`), **okka saari** `innerHTML` ki istham — okate reflow. Button meeda `data-id` stamp chestham — click appudu e product o adi cheptundi. User input ki matram ee pattern vaadaku (XSS) — `textContent` vaadu.

4.3 renderCart(details)

Three responsibilities, in order:

1. **Badge** — `badgeEl.textContent = details.count` (the reduce-computed quantity sum).
2. **Lines** — same `map().join("")` pattern; each `+/-/X` button carries **two** data attributes: `data-action` (what to do) and `data-id` (to which line). The delegated handler reads both.
3. **Summary** — subtotal, coupon row (rendered *conditionally* — the ``${details.discount > 0 ? ... : ""}` ternary drops it entirely when no coupon is active), GST, and total, all formatted through `money()`.

It receives `details` (the output of `cartDetails`) rather than raw state — the render layer never computes; it only formats what the store already derived.

4.4 `toast(message)`

```
const el = document.createElement("div");
el.className = "toast";
el.textContent = message;
zone.appendChild(el);
setTimeout(() => el.remove(), 1800);
```

How it works: create a div → set its **text** (not HTML — toasts echo coupon codes, which are user input!) → append (CSS `@keyframes pop` animates it in using only `transform` / `opacity`, the two properties that skip layout and paint) → a 1.8-second timer removes it. Multiple toasts stack because the zone is a flex column and each element is independent.

5. Section 4 — The Wiring, Function by Function

5.1 The two state variables

```
let state = store.loadState(); // cart - persisted
let view = { search: "", category: "all", sort: "default" }; // UI - not persisted
```

Two kinds of state, deliberately separated: your **cart** should survive a refresh; your **half-typed search** shouldn't. Knowing *what not to persist* is as much a design decision as what to persist.

5.2 `update(newState)` — the heartbeat

```
function update(newState) {
  state = newState;
  store.saveState(state);
  render();
}
```

Every mutation in the entire app funnels through these three lines: **assign** → **persist** → **repaint**. Because there is exactly one path, there is exactly one place to add logging, undo history, or a "saving..." indicator later. This function *is* the architecture.

Telugu: App motham lo prathi marpu ee **mudu lines** gunda ne veltundi: state assign → save → render. Okate daari undatam valla — repu logging, undo, analytics em add cheyyalanna **okka chote** add chestham. Ee chinna function e architecture.

5.3 `render()` — the visible-products pipeline

```
let visible = PRODUCTS
  .filter(p => p.name.toLowerCase().includes(view.search.toLowerCase()))
  .filter(p => view.category === "all" || p.category === view.category);
if (view.sort === "price-asc") visible = [...visible].sort((a, b) => a.price - b.price);
...
renderProducts(visible);
renderCart(store.cartDetails(state));
```

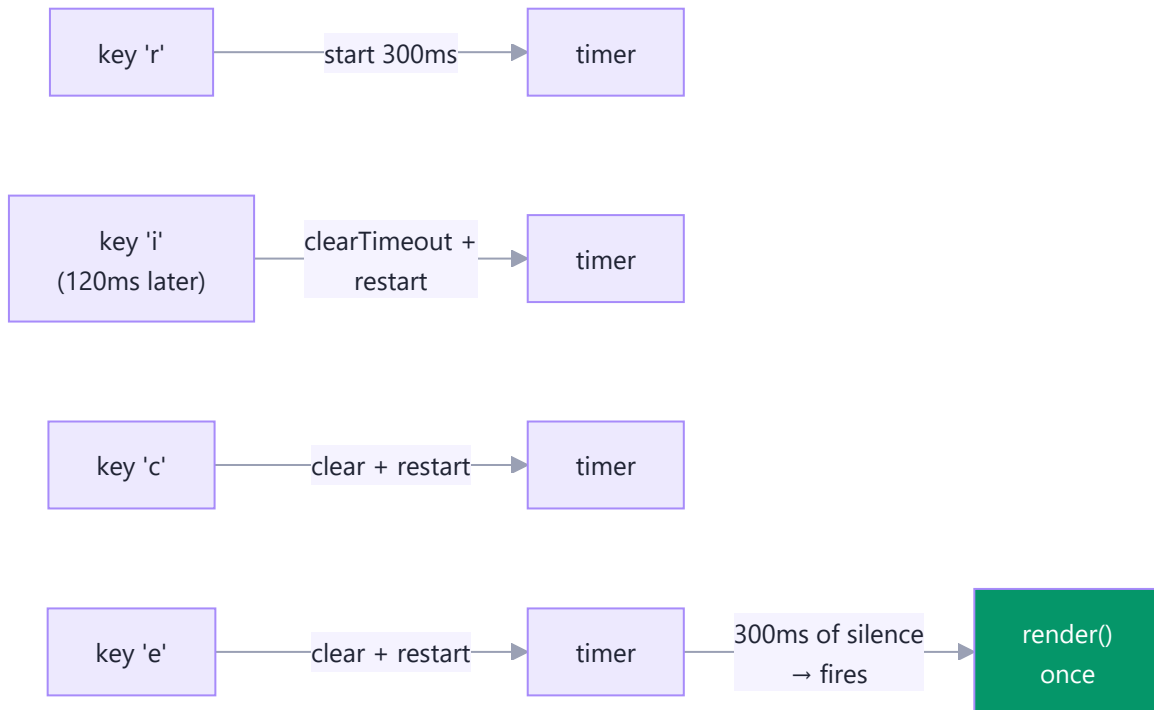
How it works:

1. Filter by search text (case-insensitive on both sides).
2. Filter by category — `"all"` short-circuits the check.
3. Sort **a copy** — `[...visible].sort(...)` — because `.sort()` mutates the array it's called on.
4. Paint both regions. The cart is re-rendered even for a search change — simpler than tracking *which* part changed, and cheap at this scale. (When that stops being cheap is exactly when a virtual DOM earns its keep.)

5.4 `debounce(fn, ms)` — the closure at work

```
const debounce = (fn, ms) => {  
  let timer;  
  return (...args) => {  
    clearTimeout(timer);  
    timer = setTimeout(() => fn(...args), ms);  
  };  
};
```

How it works: `debounce` runs **once** and returns a new function. `timer` lives in the closure — the returned function remembers it between calls. Every keystroke cancels the previous timer and starts a fresh 300ms one; only when you *pause* does the timer survive long enough to fire `render()`. Ten keystrokes → one render instead of ten.



Telugu: `timer` variable **closure** lo bathiki untundi — return ayina function daanini gurthu pettukuntundi. Prathi keystroke paatha timer ni cancel chesi kotha 300ms timer pedutundi. Nuvvu **aagithe ne** timer complete ayyi render avutundi. 10 keys → 1 render, 10 kaadu.

5.5 The seven event listeners

Listener	Element	Event	What it does
Search	<code>#search</code>	<code>input</code> (debounced)	<code>view.search = value</code> → <code>render()</code>
Category	<code>#category</code>	<code>change</code>	<code>view.category = value</code> → <code>render()</code>
Sort	<code>#sort</code>	<code>change</code>	<code>view.sort = value</code> → <code>render()</code>
Add to cart	<code>#product-grid</code>	<code>click</code> (delegated)	read <code>data-id</code> → <code>addItem</code> → <code>update</code> → toast
Cart actions	<code>#cart-items</code>	<code>click</code> (delegated)	read <code>data-action</code> + <code>data-id</code> → inc/dec/remove → <code>update</code>
Coupon	<code>#coupon-form</code>	<code>submit</code>	<code>preventDefault</code> → <code>applyCoupon</code> → <code>update</code> → toast ok/fail
Clear / toggle	<code>#clear-cart</code> , <code>#cart-toggle</code>	<code>click</code>	<code>clearCart</code> → <code>update</code> / CSS class flip

Why the two bold ones are delegated: cart buttons are destroyed and recreated on every render — listeners attached to them would die each time. The *container* survives renders, so its single listener works forever, including for buttons that don't exist yet. `e.target.closest("button[data-action]")` walks up from wherever the click landed to the button (or returns `null`, in which case we exit).

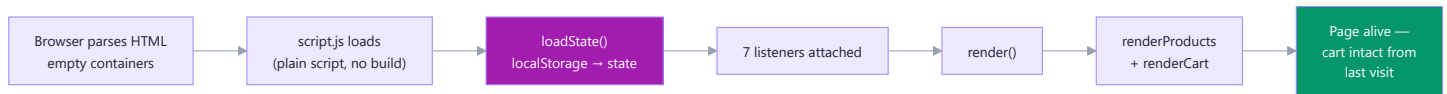
The action-table trick in the cart handler:

```
const actions = {
  inc:    () => store.setQty(state, id, qty + 1),
  dec:    () => store.setQty(state, id, qty - 1),
  remove: () => store.removeItem(state, id),
};
update(actions[btn.dataset.action]());
```

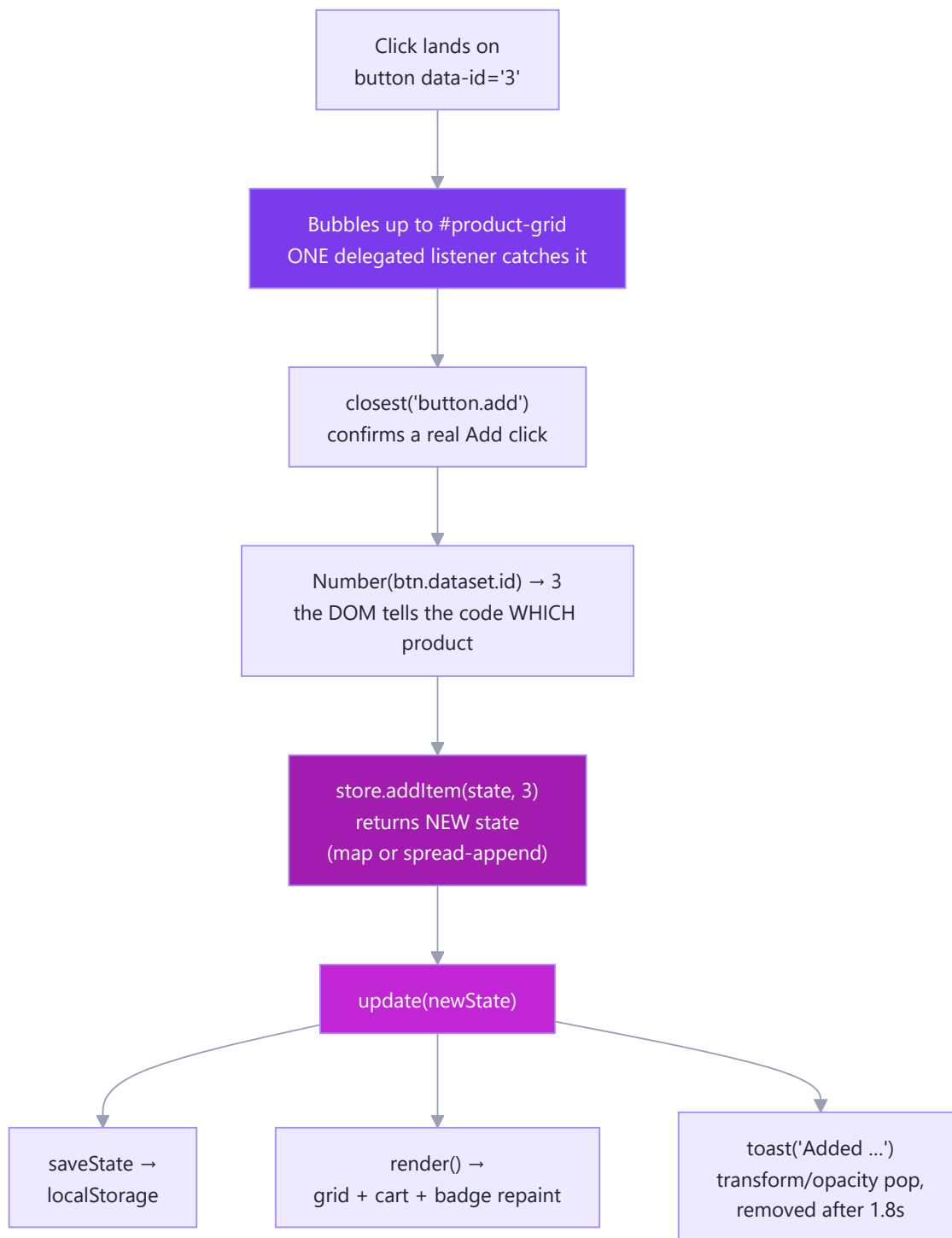
An object lookup replaces an if/else ladder — adding a "move to wishlist" action tomorrow is one new line, not a new branch.

6. The Flows — What Happens When...

6.1 Page load



6.2 "Add to Cart" click — the full journey



6.3 Money update — from click to new total

Every price change on screen is this chain — no shortcuts, no cached numbers:



6.4 Quantity – on the last unit

`dec` → `setQty(state, id, 0)` → `qty < 1` → `removeItem` → line vanishes from the next render. The – button never knew it was the last unit; the *store rule* handled it. That's the payoff of putting logic in one place.

6.5 Coupon submit

Enter or button → `submit` event → `preventDefault()` (stops the browser's default full-page reload) → `applyCoupon` normalises and validates → `{ state, ok }` → `update` + a success/fail toast. Invalid codes change nothing: the same state object comes back.

Companion to [README.md](#). Part of the 50-Day Challenge — the cart build from the JS practice lab, explained to the last function.